BhumiSetu | ভূমিসেতু

LandTech 2026 concept note · P1 · P2 · P3 · P5

BhumiSetu connects four land workflows: digitizing records (P1), verifying deeds (P2), detecting conflicting transactions (P3), and helping citizens check before buying (P5).

The human idea: A buyer should see a conflict in the records before it becomes a conflict between families. BhumiSetu turns scattered papers into reviewed digital evidence, explains contradictions in plain language, and leaves the decision with an officer.

1. The problem in Bangladesh

For many families, a plot is their largest savings. Yet the evidence for one parcel can be spread across CS, SA, RS and BS khatians, porchas, registered deeds and later mutation (namjari) entries. A deed recorded at a sub-registry office and a mutation record can describe different parties or areas; older and newer survey identifiers may also need careful reconciliation. Someone checking only one paper can miss another transfer of the same land. The Ministry of Land's e-Mutation process routes mismatched details for officer review, which is the right place for our evidence alerts.

One broken link in that chain can hide a second sale or create uncertainty for an honest buyer. A warning is evidence for review, not a finding of fraud or a declaration of legal title.

2. Four problems addressed in one working system

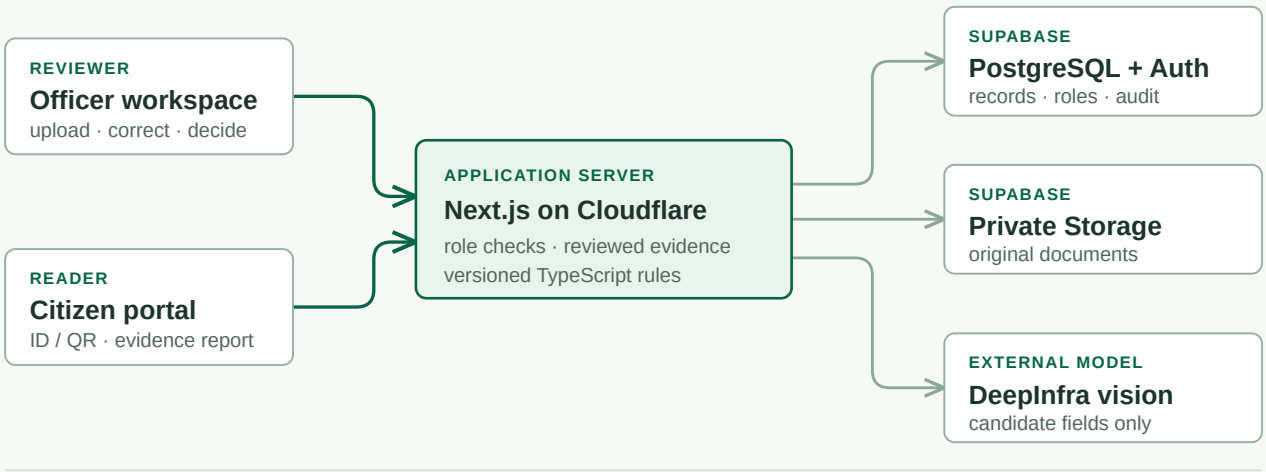
Challenge	What BhumiSetu already does
P1 · Digitize old records	Officers upload Khatian, Porcha, Deed and Mutation PDFs, DOCX files or images. Text/vision extraction proposes searchable fields; officers inspect the original and edit the result before verification.
P2 · Verify deeds & ownership evidence	A deed image, document or QR payload is checked against a reviewed record. Field-by-field comparisons show matched, mismatched or inconclusive results; enrolled originals also have SHA-256 copy hashes.
P3 · Catch fraud & double-selling	Versioned rules flag overlapping or excess sales, ownership-chain gaps, deed–mutation conflicts and suspicious dates, with the source documents behind each alert.
P5 · Check before buying	A citizen enters a land ID or QR link to see the published parcel's recorded parties, area, history and evidence report in Bangla or English, then print a dossier.

3. One parcel, two users, one evidence trail

An officer uploads the source papers, reviews AI-suggested fields, records corrections with reasons and publishes verified evidence. A buyer searches the parcel or scans its QR; the same reviewed data powers a readable conflict report and deed comparison. The officer sees the underlying documents and rule trace before deciding a related mutation case.

Illustrative synthetic case: A parcel has 20 decimals recorded, while two deeds from the same seller claim 12 decimals each. BhumiSetu shows the 24-versus-20 discrepancy and both deeds before a buyer pays or an officer approves a mutation.

4. Solution architecture and tech stack



Officer-reviewed evidence reaches a versioned rule engine through protected server routes.

Layer	Current implementation
Citizen and officer interface	Next.js 15 App Router, React 18, TypeScript, Tailwind; Bangla/English catalogues; land-ID and QR journeys
Application and deployment	Next.js server routes packaged by OpenNext for Cloudflare Workers; server-side role checks and protected API actions
Records and access	Supabase Auth, PostgreSQL, row-level security, private Supabase Storage; original file hash and audited edits
Document understanding	PDF.js, DOCX parser, image preparation, DeepInfra-hosted Qwen vision model; officer reviews every extracted field
Verification	Versioned TypeScript rules and deed field comparison; evidence-linked reports and mutation review

Human control is part of the architecture. AI proposes document fields; deterministic rules produce the conflict indicator. Officers compare extraction with the source, verify documents and decide mutation cases. Original files stay private; published citizen views use reviewed data.

5. Why the four parts fit together

P1 creates the reviewed evidence; P2 checks a presented deed against it; P3 finds contradictions across records; P5 gives the buyer an understandable view of the result. These are linked product journeys, not four disconnected demos. Corrections preserve old/new values and reasons in an officer audit trail, while the server blocks approval of unresolved conflict cases.

6. From prototype to nationwide use

Phase 1 — district pilot. Start with a controlled, synthetic-to-authorized-data transition in one district. Define a parcel matching key using district, upazila, mouza, JL, dag, khatian and survey type; map CS/SA/RS/BS changes without collapsing distinct historical identifiers. Have officers validate ambiguous matches and measure extraction corrections, alert precision and time saved.

Phase 2 — interoperable exchange. With government authorization, connect registry, land-record and mutation workflows through documented import/API contracts. Preserve each source's issuing office, document date, version and hash. Route conflicting updates to a review queue instead of silently replacing one record with another.

Phase 3 — national operations. Scale ingestion in batches by district, monitor processing and audit events, and use role-scoped access so only authorized officers can alter reviewed data. Continue human review for disputed ownership and publish only the minimum citizen-facing evidence needed for informed checking.

7. Why this matters

BhumiSetu turns a pile of papers into a question a person can act on: **“Do these records agree about this parcel?”** A buyer gets a timely reason to pause; an officer gets the documents and rule trace needed to investigate. The prototype demonstrates all four connected challenges with invented records and original sample files, without government-site scraping or real citizen data.

Live prototype: https://bhumisetu.adribmahmud.workers.dev
Code and reproducibility: https://github.com/AdribMahmud101/bhumisetu

8. Team (four required members)

Role	Name / affiliation
Student 1 · Team Leader (DIU)	**[TEAM LEADER NAME / ID]**
Student 2 (DIU)	**[STUDENT NAME / ID]**
Faculty Mentor	**[FACULTY MENTOR NAME / DEPARTMENT]**
Industry Mentor	**[INDUSTRY MENTOR NAME / ORGANIZATION]**

Team details are placeholders until confirmed by the applicants.

Land context: Ministry of Land e-Mutation steps · Ministry of Land portal